



Architect4Hire

DELIVERY & INVESTMENT PLAN

Event-Driven Order Fulfillment

Reference Build & Architecture Baseline

Prepared for Architect4Hire Demo Client · **Robert Felkins**



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Workstreams

#	Workstream	Purpose	Complexity	Effort	Skill set
1	Solution scaffold & Contracts	Aspire AppHost + emulators, ServiceDefaults, shared Contracts	High	2 d	Architect
2	Order saga & compensation	Saga host, event log, read model, compensation on every failure path	High	3 d	Architect
3	Reacting services	Inventory (concurrency), Payment (idempotent), Fulfillment, Notification	Medium	4 d	Architect
4	Resilience & observability	Retry/backoff, dead-lettering, idempotency, distributed tracing	Medium	2 d	Architect
5	Angular frontend	Customer order status view + operations view over the read model	Medium	2 d	Architect
6	Architecture baseline & ADRs	Design doc, C4 + sequence diagrams, ADRs, SCRUB prompt library	Low	2 d	Architect

Effort is first-pass; sequencing and estimates firm up in Vision. Delivered solo — architect-only, no development squad.

Correctness under failure



Partial-failure inconsistency

CRITICAL

A multi-step order can fail mid-flow. Without compensation, a failed step leaves stock reserved or a charge stuck.



Duplicate-delivery risk

CRITICAL

At-least-once delivery means a payment callback can arrive twice. Without idempotency, that is a double charge.



Oversell under contention

IMPORTANT

Two buyers racing for the last unit can both succeed unless inventory reservation is concurrency-correct.



No durable baseline

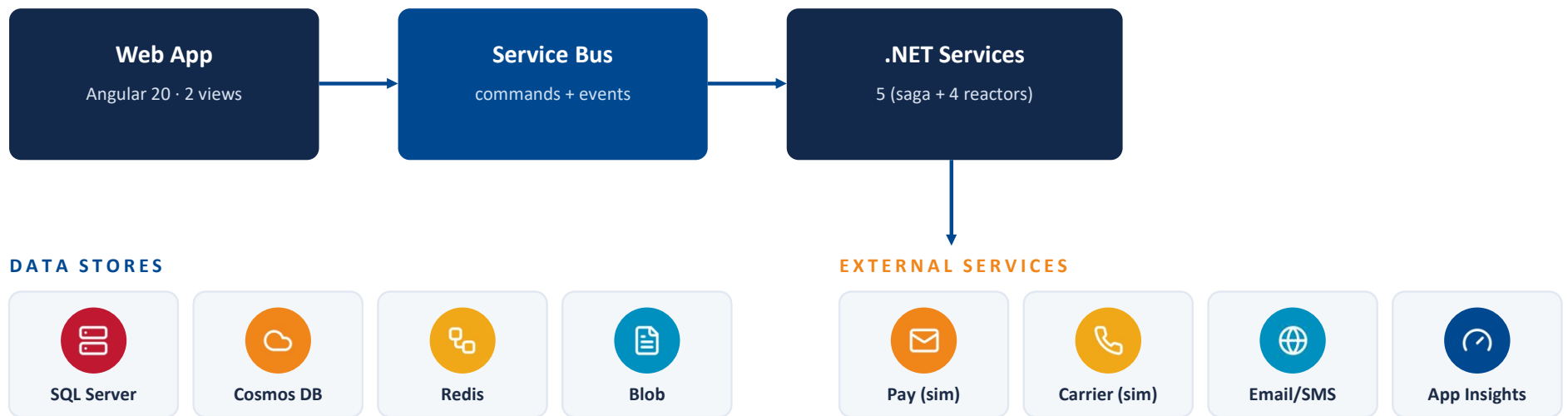
IMPORTANT

Without ADRs and a documented baseline, the design and the code drift apart the moment the build ships.

Benefits & measures

Benefit	Level	Description	Measure
Correctness under failure	High	Every failure path compensates — no order left with stock reserved or a charge captured	Partial-order incidents -> 0
Idempotency	High	Duplicate payment callbacks are no-ops; poison messages dead-letter, not loop	Double-charge incidents -> 0; poison isolated
Concurrency safety	Medium	Concurrency-correct inventory; status read model converges under one second	Oversell incidents -> 0
Durable baseline	Medium	Design doc, ADRs, C4 diagrams, and SCRUB library as a durable source of truth	Onboarding & handoff time down

The OrderFlow landscape

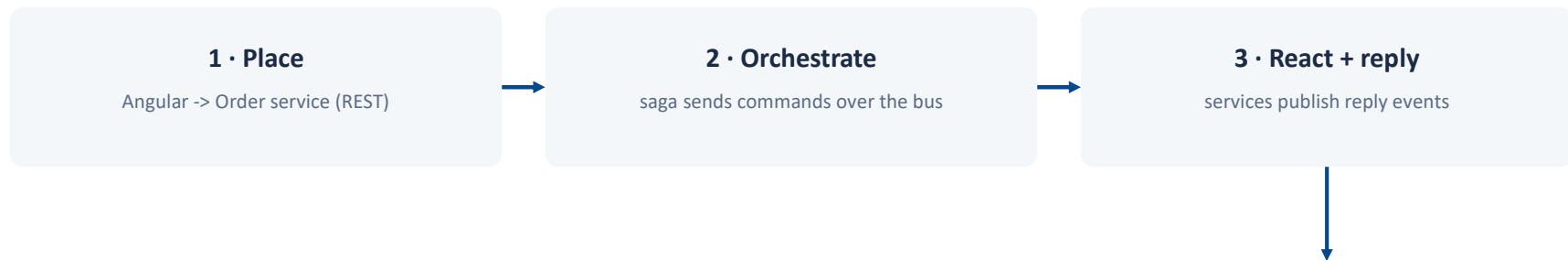


Orchestrated by a saga: the Angular app calls the Order service; commands and events flow over Service Bus. Cosmos holds the event log, Redis the status read model, SQL the inventory + payment records — all on Aspire-local emulators.

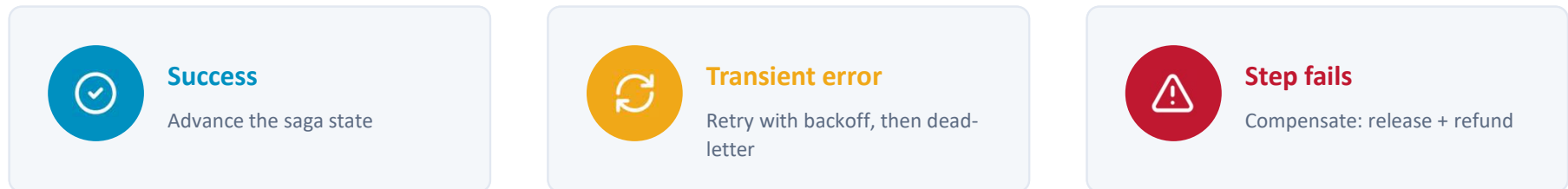
HOW IT RUNS



Order saga lifecycle

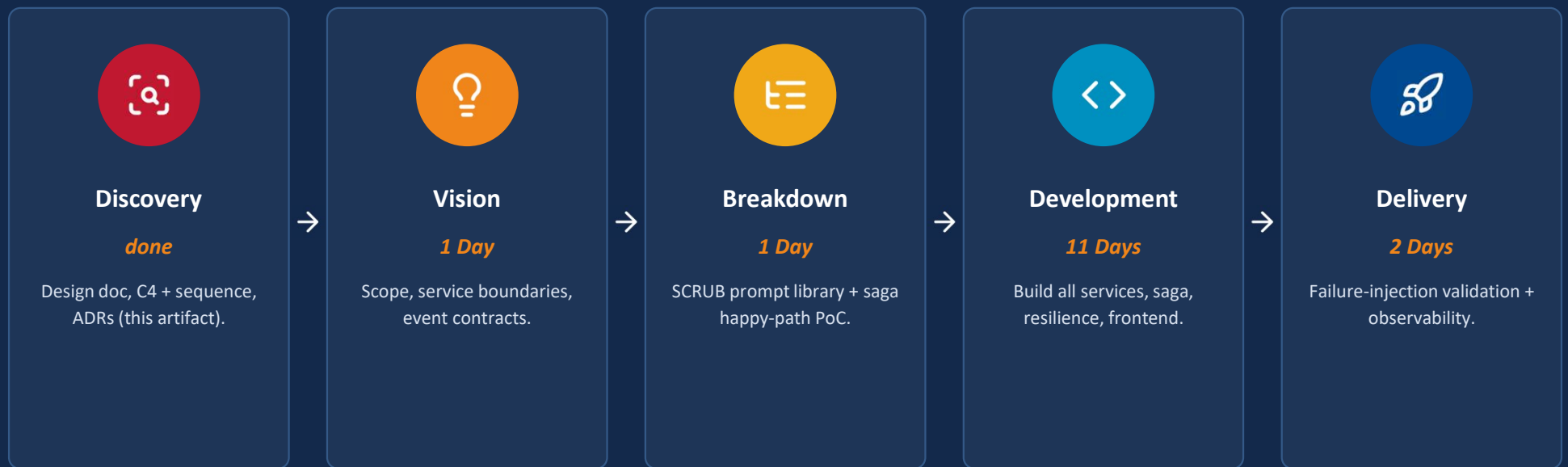


FAILURE HANDLING — BY DESIGN



Every failure path compensates: a declined payment releases the reservation; a failed dispatch refunds and releases. No order is left partial.

Delivery roadmap



~6 weeks end to end · architect-only, no squad

Labor estimates

Phase	Sprint allocation (rate)	Notes
Vision	Architect: 8h @ \$75/h	Scope, service boundaries, event contracts
Breakdown	Architect: 16h @ \$75/h	SCRUB prompt library + saga happy-path PoC
Development	Architect: 88h @ \$75/h	Build all services, saga, resilience, frontend
Delivery	Architect: 16h @ \$75/h	Failure-injection validation + ADRs + handoff
Total	Architect: 128 hours / \$9,600	~\$9,600

Investment expectations

Category	Initial investment	Ongoing investment
Software	No additional Azure spend	No additional Azure spend
Hardware	No additional Azure spend	No additional Azure spend
Services	No additional spend	No additional spend
Architect4Hire labor	~\$9,600 over ~6 weeks	No additional spend
Client labor	~4h / week of an SME	~1h / week
Total	~\$9,600	Minimal



No cloud spend to build — the system runs on Aspire-local emulators against real Azure SDKs. The spend buys senior judgment and a working reference architecture, not new resources.

Total cost of ownership

Horizon	Cost	Rationale
Initial investment	\$9,600	OrderFlow build + architecture baseline
Year 1	opt.	Optional fractional support @ \$75/h as needed
Year 2	\$0	No new spend
Year 3	\$0	No new spend
Year 4	\$0	No new spend
Year 5	\$0	No new spend
5-year TCO	\$9,600	One-time build; ongoing support optional



The build is one-time; the artifacts compound. The design doc, ADRs, and SCRUB library keep paying off long after roll-off — ongoing fractional support is optional, not required.

AGREEMENT



Review, approval & document control

REVIEW & APPROVAL

Role	Name	Reviewed	Agreed
Client Sponsor	Architect4Hire Demo Client	[Y]	[Y]
Architect	Robert Felkins	Yes	Yes

DOCUMENT CONTROL

Rev	Author	Date	Description
1	Robert Felkins	7/13/2026	Initial draft — reference build plan
[2]	[author]	[date]	[change]

This document is versioned and signed off before the build phases begin.



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LET'S TALK

Questions?

Push on the findings, the sequence, the estimates — that's what this conversation is for.



robert@architect4hire.com



(312) 493-9696



architect4hire.com

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